

Thermal conductivity inside prismatic lithium-ion cells with dependencies on temperature and external compression pressure

Marco Steinhardt*, Elisabeth Irene Gillich, Maximilian Stiegler, Andreas Jossen

Technical University of Munich (TUM), Institute for Electrical Energy Storage Technology (EES), Arcisstrasse 21, Munich, 80333, Germany

ARTICLE INFO

Keywords:

Li-ion battery
Thermal conductivity
Thermal contact resistance
Thermal resistance

ABSTRACT

Significant temperature increases and gradients inside lithium-ion battery cells are highly likely during extreme fast charging and discharging, with adverse effect on aging and safety. To assess the temperature inside the cell, the thermal conductivity of the electrode-separator composite must be determined precisely. Furthermore, the dependency of the thermal conductivity on battery states like the cell temperature or composite layer pressure has to be considered. In this work, these dependencies are investigated for a large-format lithium-ion cell with a flat-wound jelly roll and prismatic aluminum hardcase with a Nickel Manganese Cobalt (NMC) cathode and graphite anode. A precise thermal conductivity test bench with defined compression boundaries for the battery cell is constructed in compliance with the guarded heater principle. Validation tests with a stainless steel reference material prove a high accuracy and robustness against changes in compression pressure. In addition, a novel method to obtain the thermal conductivity of the aluminum cell-case is introduced. According to the measurement result, the thermal conductivity increases by 11.9% when the compression pressure rises by 37.1 kPa, which is explained by thermal contact resistances. Compared to the absolute value at 20 °C, the thermal conductivity decreases with increasing temperature by more than -1% per K.

1. Introduction

Several car manufacturers have started to mass-produce electric vehicles with lithium-ion batteries as a result of stricter political CO₂ emission targets. From the perspective of the consumer, shorter driving range and longer charging time still remain major disadvantages compared to cars with internal combustion engines [1]. Extreme fast charging with an 80% change in state of charge (SoC) within 10 min constitutes a major thermal challenge for lithium-ion cells and their thermal management systems [2]. Furthermore, cell design strategies aiming at higher energy densities are often accompanied by higher heat losses for equal charging power [1]. This represents a trade-off between fast charging and energy density due to thermal aspects.

Temperature rise and spatial temperature gradients inside the cell are among the main thermal challenges during fast charging. Most safety devices in lithium-ion cells are designed for an upper operational temperature of 60 °C [3] to reduce the risk of thermal runaway events with onset temperatures starting from 80 °C [4]. Waldmann et al. [5] examined the relationship between C-rate and temperature rise with values up to 15.7 °C·C⁻¹. This emphasizes the need for active cooling if C-rates of 6 C are demanded. Thermal gradients inside the cell tend to rise with equal magnitude of 3.19 °C·C⁻¹, which was measured by

Waldmann et al. [6] for a 18650 cylindrical cell in radial direction. These gradients can reduce the lifetime of the cell [7], or can be a safety issue if the margin between 60 and 80 °C is exceeded. Temperature rise and gradients are both affected by the thermal conductivity inside the cell. Hence, thermal conductivity proves to be a decisive parameter in the area of fast charging.

The thermal conductivity inside lithium-ion cells is anisotropic due to the layered structure of the separator electrode composite. This composite is often reduced to a uniform medium with a through-plane $k_{\perp}$ and an in-plane $k_{\parallel}$ thermal conductivity. The through-plane direction represents the major heat flux direction for most prismatic hardcase cells [8]. In contrast to $k_{\parallel}$, $k_{\perp}$ includes thermal contact resistances between the layers [9], which are dependent on the mechanical boundary conditions. Vishwakarma et al. [10] claim that $k_{\perp}$ cannot be calculated by thermal conductivity values measured on material level if thermal contact resistances remain at a significantly high level. By implication, $k_{\perp}$ has to be measured for every cell type and boundary condition on stack or cell level if accurate modeling accuracy is important.

If high thermal modeling accuracy is important, dependencies on battery states like temperature or SoC have to be considered as well. For the temperature dependency of $k_{\perp}$, all available measurement data for

* Corresponding author.

E-mail address: marco.steinhardt@tum.de (M. Steinhardt).

<https://doi.org/10.1016/j.est.2020.101680>

Received 18 April 2020; Received in revised form 28 June 2020; Accepted 15 July 2020

Available online 17 August 2020

2352-152X/ © 2020 Elsevier Ltd. All rights reserved.

Table 1

Reported temperature dependency (dk/dT)/ k_{20} of the thermal conductivity in through-plane direction of the electrode-separator-composite k_L . Additional information is given on the cell case, external compression pressure during measurement, electrode structure and material composition.

Authors	Case & pressure	Electrode geometry	Anode	Ψ_A	d_A/d_{Cu}	Cathode	Ψ_C	d_C/d_{Al}	k_{20}	(dk/dT)/ k_{20}
Bazinski et al. [11]	Pouch foil; +0.5 bar	Stack	Graphite	n/a	n/a	LCO	n/a	n/a	0.35 W m ⁻¹ K ⁻¹	0.0%K ⁻¹
Werner et al. [8]	Prismatic Steel; 0 bar	Jelly Roll	Graphite	48%	8.5	LCO	29%	4.0	1.10 W m ⁻¹ K ⁻¹	-0.9%K ⁻¹
Sheng et al. [12]	Prismatic Al 3003; -0.9 bar	Stack	Graphite	n/a	0.3	LFP	n/a	0.8	0.54 W m ⁻¹ K ⁻¹	+0.1%K ⁻¹
Sheng et al. [12]	Al-plastic film; -0.9 bar	Stack	Graphite	n/a	n/a	LFP	n/a	n/a	0.52 W m ⁻¹ K ⁻¹	+0.1%K ⁻¹
Sheng et al. [12]	Al-plastic film; -0.9 bar	Stack	Graphite	n/a	n/a	NMC	n/a	n/a	0.66 W m ⁻¹ K ⁻¹	+0.1%K ⁻¹

double-coated unit cells is listed in Table 1. In this work, the relative temperature dependence (dk/dT)/ k_{20} is defined by the slope of the linear regression dk/dT divided by the absolute value k_{20} at 20 °C. Bazinski et al. [11] performed the first measurements on a 14 Ah pouch cell with LFP cathode showing no significant dependency on temperature. Two years later, Werner et al. [8] measured a significant decline with increasing temperature by -0.91%K⁻¹ on a 6.8 Ah lithium cobalt oxide (LCO) prismatic hardcase cell. This finding was supported by measurements at layer level, showing that this bottom-up approach correlated closely with full cell measurements. In 2019, Sheng et al. [12] observed a marginal increase of +0.1%K⁻¹ for three self-made battery cells composed of commercial NMC and LFP cell cores embedded in either a hardcase or an aluminum-plastic pouch foil.

Comparing (dk/dT)/ k_{20} in Table 1 reveals a strong disagreement between the result found by Werner et al. [8] those of Bazinski et al. [11] and Sheng et al. [12]. This discrepancy could be caused by different thermal contact resistances that result from the case material or external compression pressures. Measurements by Sheng et al. [12] on LFP cathodes discard any dominant effect of the cell case, while the external pressure may yield different findings on (dk/dT)/ k_{20} . The electrode geometry serves as another explanation, while Werner et al. [8] remains the only investigation with a flat-wound jelly roll geometry.

According to Loges et al. [13], (dk/dT)/ k_{20} is influenced by the electrode structure, the active material composition and thickness ratio of anode d_A and cathode d_C coating to their copper d_{Cu} and aluminum d_{Al} current collectors. Comparing the cathode coatings in Table 1 draws the conclusion that LCO cathodes exhibit a completely different temperature dependency than NMC or LFP. Werner et al. [8] is the only investigation that states porosity values for anode Ψ_A and cathode Ψ_C , which discard this parameter for any explanations. Finally, the significant difference in thickness ratios between Werner et al. [8] and Sheng et al. [12] is probably another reason for different (dk/dT)/ k_{20} values.

This work aims to clarify some disagreements in literature on (dk/dT)/ k_{20} by repeating the work of Werner et al. [8] on cell level with an NMC cathode coating. Furthermore, the dependency on external compression pressure is part of the scope for evaluating the influence of thermal contact resistances, which did not form part of any investigation in Table 1.

The remainder of this paper is organized in three sections. In the experimental Section 2, the cell being tested and the measurement technique based on the guarded heater principle is described and validated with a standard reference material. Furthermore, thermal conductivity tests are shown at cell level with different temperatures and external pressure. All necessary information for transforming the thermal conductivity at cell level to the unit cell level is given in the theory and modeling Section 3. The unit cell in this paper is defined as a cube consisting of the smallest repetition of the cell layers keeping conform with the definition of Bandhauer et al. [14]. An overview of the entire transformation process is given in Fig. 7 a). The last Section 4 combines the experimental and theoretical results and discusses the temperature and pressure dependency with the findings in Table 1.

2. Experimental

2.1. Cell and reference material

The cell being tested is a prismatic hardcase PHEV-2 cell with a nominal capacity of 34 Ah, a graphite anode and NMC-111 (Li(Ni_{1/3}Co_{1/3}Mn_{1/3})O₂) cathode. The cell has been manufactured by VARTA Microbattery GmbH (Germany) for research purposes and is equipped with an internal Type K thermocouple inserted into the center of the jelly roll and one reference electrode. The maximum continuous C-rate for charge and discharge is specified by 2 C. An electrical resistance of 1.5 mΩ is rated for a 10 s current pulse response. The cell in question has never been operated before this investigation and has been stored at a SoC of 15%.

Fig. 1 a) and b) shows a computer-aided design (CAD) drawing and photograph of the cell that contains a single vertical flat wound jelly roll. Fig. 1 c) and d) indicate length specifications of the jelly roll and the cell case, which are important for the conductivity calculations in subsequent sections. The case, with thicknesses ($d_{c,b}$, $d_{c,s}$) ranging between 0.7 and 1.5 mm, has no physical contact with the jelly roll except for the two largest surfaces. This contact area is marked in Fig. 1 c) by the red rectangular at a height $d_{j,r,h}$ of 82 mm and a length d_s of 117 mm. The case is made of AW-3003 aluminum alloy (AlMn1Cu) with a 1.1% manganese composition. This alloy is a common case material for prismatic battery cells [14] and its thermal conductivity ranges between 160 and 190 Wm⁻¹K⁻¹ [15].

Table 2 lists geometrical parameters of the unit cell layers consisting of two double coated electrodes, two separator layers and one layer for each current collector (CC). Accordingly, the ratio of coating to current collector thickness for this cell is 4.3 for d_C/d_{Al} and 6.0 for d_A/d_{Cu} , which are both in the range reported by Loges et al. [13]. The heights and lengths listed in Table 2 can be used to calculate the ratio of anode overhang. Dividing their areas results in an anode overhang of 9.8%, which corresponds to the value published by Lewerenz et al. [16].

To evaluate the measurement error of the test bench introduced in the following subsection, a suitable reference material had to be selected. Details of this procedure following the criteria defined by Hust et al. [17] can be found in Appendix A. In result of this, a cuboid (150x25x91 mm) made of 1.4301 (X5CrNi18-10) stainless steel was purchased for quantifying the measurement error in this study. The reference material is shown in Fig. 1 a) (tag 6).

2.2. Test bench

The thermal conductivity test bench is based on the single-specimen guarded hot plate method, which is one of the most accurate steady-state measurement techniques [18]. Fig. 2 a) shows an overview of the whole test bench placed in a climate chamber that provides a stream of air with defined temperature T_{air} . The test object is stacked with other components between two aluminum plates, while the right plate (tag 2) serves as a heat sink containing a water-glycol (50:50) coolant with the temperature $T_{coolant}$. An equal temperature for T_{air} and $T_{coolant}$ is used at all times, to ensure that heat losses from the component stack to the environment remain equal. To keep radial heat losses to a minimum, a

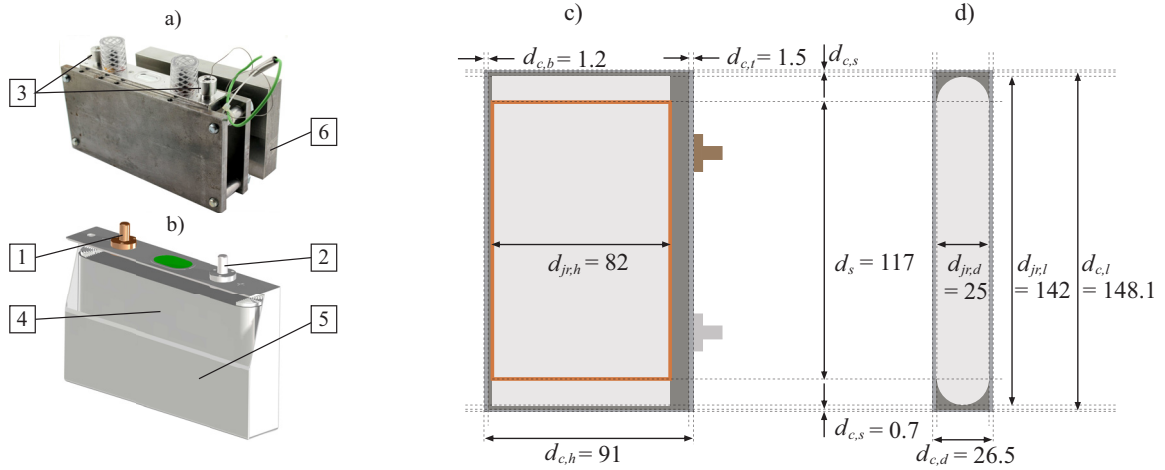


Fig. 1. Photograph a) and CAD drawing b) of reference material and cell being tested with corresponding vertical c) and horizontal d) section planes. Length specification are given in mm for the jelly roll and prismatic hardcase. 1: Negative terminal. 2: Positive terminal. 3: Feed-through for internal sensors. 4: Jelly roll. 5: Case. 6: Stainless steel reference material.

Table 2

Specification of the electrode-separator-composite given by the cell manufacturer. Length and height of the separator have not been specified.

Unit cell layer	Height / mm	Thickness / μm	Porosity / %	Length / m
2x NMC 111	74.0	64.0	27	10.9
1x Al CC	74.5	15.0	0	10.9
2x separator	n/a	21.6	42	n/a
2x graphite	77.0	59.5	30	11.5
1x Cu CC	77.5	10.0	0	11.5

4 cm thick isolation made of Styrodur® 3000 CS from BASF (tag 3) is wrapped around the stack with a thermal conductivity of $0.033 \text{ W m}^{-1} \text{ K}^{-1}$. With a maximum temperature difference between the test object and T_{air} of 10.366 K during all measurements, lateral heat losses are smaller than 0.121 W.

A defined force F_{spring} can be applied to the stack by four springs (tag 1) with a spring constant of 47.734 N mm^{-1} . In the following experiments, the cumulative force of the springs ($4 \cdot F_{\text{spring}}$) was always set to 500 or 1000 N. This translates to a compression pressure of 36.6 and 73.3 kPa for the reference material and 37.1 and 74.2 kPa for the cell being tested. The difference in compression pressure between the reference material and the cell is caused by different cross section areas specified in subsection 2.1.

Fig. 2 b) shows details on the measurement stack consisting of a guarded heater (tag 4), the main heater (tag 5), the test object (tag 6) and a heat flow equalizer (tag 7) that provides a uniform heat flow for the test object. The main heater generates the heat flux $\dot{Q}$ for measuring thermal conductivity, which passes through the test object, the equalizer and finally sinks on the right aluminum plate. To assure equal and low contact resistances, Keratherm KP 92 thermal grease made by Kerafol with a thermal conductivity of $10 \text{ W m}^{-1} \text{ K}^{-1}$ is used between any contact surface passed by $\dot{Q}$.

A guarded heater (tag 4) is deployed to minimize heat losses on the left side of the main heater. Both heaters consists of a heating wire, which is paste between Styrodur® isolation and a copper plate with a thickness of 1 cm. The copper plates equalize the concentrated heat flux of the heating wires while the Styrodur® directs the heat flux towards the test object. Two Pt100 temperature sensors are installed to measure the temperature of the guarded heater copper plate T_{GH} and the main heating wire T_{H} .

The linear power supply GPD 3303D from GW-Instek supplies the electrical power for both heaters. The main heater operates in constant current mode I_{H} and the corresponding voltage drop U_{H} is measured with separated sense wires by the NI-9228 card made by National Instruments. The current of the guarded heater I_{GH} is automatically adjusted to reduce the difference between T_{H} and T_{GH} , which never exceeded $\pm 0.0467 \text{ K}$. Hence, heat losses at the backside of the heater

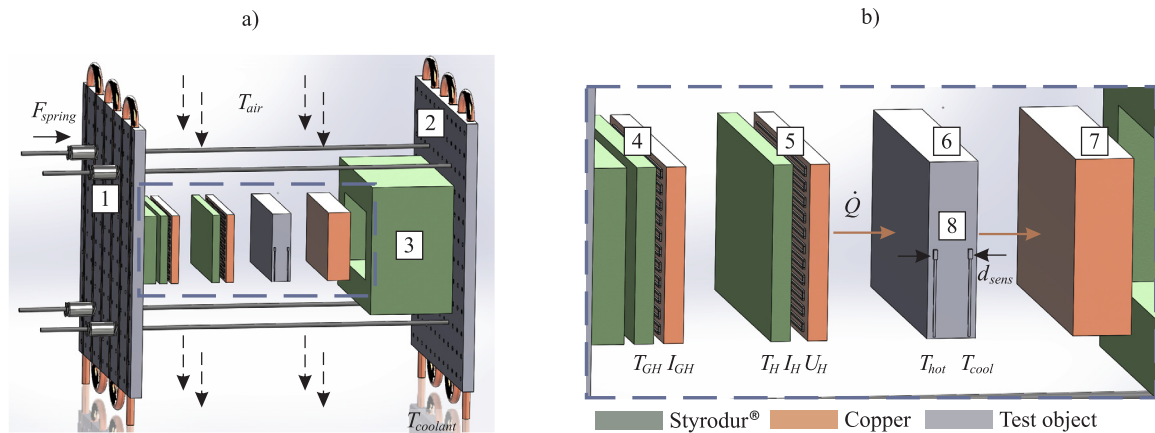


Fig. 2. Overview a) and details b) of the thermal conductivity test bench for prismatic battery cells used in this work. 1: Four springs for defined compression force F_{spring} . 2: Heat exchanger serving as heat sink. 3: Radial heat isolation. 4: Guarded heater. 5: Main heater generating heat flow $\dot{Q}$. 6: Test object (cell or reference material). 7: Heat flow equalizer. 8: Temperature sensors for test object.

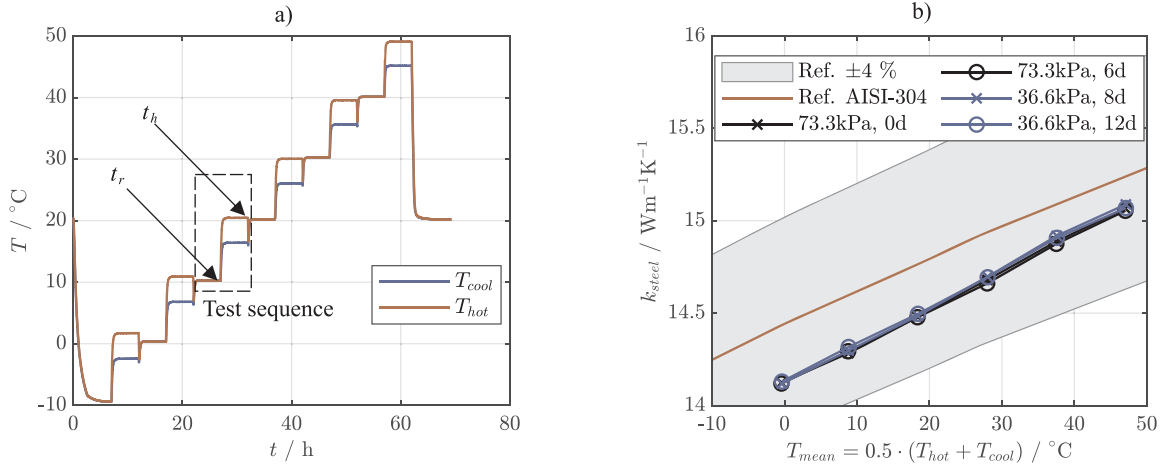


Fig. 3. a) Temperature readings during test bench validation with a stainless steel reference sample showing the evaluation point during the heating phase t_h and the rest phase t_r for one test sequence. b) Test bench validation with AISI-304 reference [19] at two compression pressures with four measurement taken between day 0 and 12.

are never greater than 2.1 mW.

The temperature gradient ($T_{hot} - T_{cool}$) in Fig. 2 is measured by two DIN EN 60751 conform 100 Ω platinum-chip temperature sensors (JUMO PCA 1.2005.1L F0,15). Both sensors are attached on the lateral surface of the test object at a distance of d_{sens} . Acquiring the temperature gradient at the lateral surface brings the advantage of not being influenced by thermal contact resistances compared to the acquisition at the two largest object surfaces. Before the sensors were attached, readings were calibrated between -10 and 40 °C with a Fluke Calibration 1524 Thermometer and a FLUKE-TEMP-5622-05-P temperature sensor with an accuracy of 0.04 °C. After the calibration, the deviation of T_{hot} and T_{cool} from the reference never exceeded 0.013 °C. In general, all temperature sensors are acquired by the National Instruments module NI-9261 in four-wire high-resolution mode, with a sample rate of 1 Hz.

2.3. Test validation

Fig. 3 a) shows the general test procedure used for any thermal conductivity measurement in this work by the temperature readings of T_{cool} and T_{hot} . The whole measurement is split into six test sequences for six different set temperatures between -10 and 40 °C. Every test

sequence contains a rest and a heating phase. At the beginning of the rest phase, $T_{coolant}$ and T_{air} is set to the desired temperature. Effects arising from the temperature history are eliminated by this rest phase and measurement errors can be analyzed and calibrated at its end t_r . After the rest phase, the heating phase starts with equal duration. At the end of the heating phase t_h , the heating power is switched off automatically and $T_{coolant}$ and T_{air} are adjusted to a new set point.

Eq. 1 is always used for calculating the measured thermal conductivity k_{mes} . The constant electrical current of the main heater I_H and the measured heater voltage U_H are multiplied representing the heat flux of the main heater $\dot{Q}$. The parameter d_{sens} is the distance between two temperature sensors (see Fig. 2 b)) and A is the cross-section area of the test object orthogonal to the heat flow. The temperature difference ΔT_{h-c} between T_{hot} and T_{cool} is measured at the end of the rest t_r and heating phase t_h . The subtraction of $\Delta T_{h-c}(t_r)$ from $\Delta T_{h-c}(t_h)$ constitutes an online calibration with the assumption that any temperature difference at t_r is only caused by measurement errors. By this online calibration, long time drifts and nonlinear uncertainties can be reduced.

$$k_{mes} = \frac{I_H \cdot U_H \cdot d_{sens}}{A \cdot (\Delta T_{h-c}(t_h) - \Delta T_{h-c}(t_r))} \quad (1)$$

Fig. 3 a) shows the temperature readings for the reference material

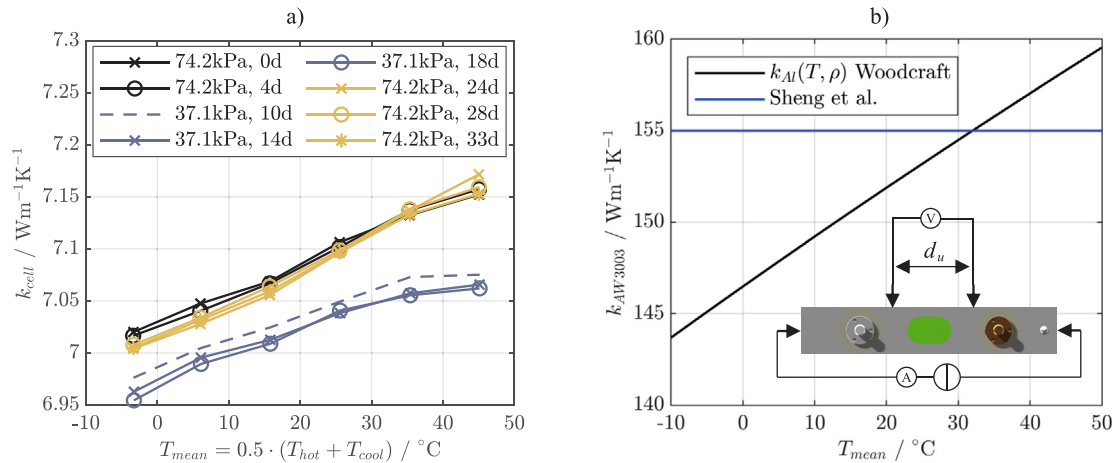


Fig. 4. a) Thermal conductivity of the cell k_{cell} at different pressures between 37.1 and 74.2 kPa with the first measurement started at day 0 and the last at day 33. b) Thermal conductivity of the aluminum alloy k_{AW3003} , which is used for the case conductivity k_{case} by Sheng et al. [12] (blue line) and in this work (black line). In this work, k_{AW3003} is calculated by electrical resistivity measurements ρ and the procedure developed by Woodcraft [22] $k_A(T, \rho)$. The setup for the electrical resistivity measurements comprises one current source and one voltage sensing in the middle part of the cell. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
Chemical composition of stainless steel standards.

Component	1.4301 [20]	AISI-304 [19]
C / %	0.07	0.08
Mn / %	2.00	2.00
Si / %	1.00	1.00
Cr / %	17.5 - 19.5	18.0 - 20.0
Ni / %	8 - 10.5	8 - 12

introduced in Subsection 2.1, with I_H set to 2.11 A, a measured heater voltage U_H between 22.519 and 22.549 V at t_h and a temperature sensor distance d_{sens} of 17.04 mm. $T_{coolant}$ and T_{air} were varied between -10 and 40 °C in increments of 10 °C. The duration of the rest and heating period was set to 5 h, resulting in a total measuring time of 60 h. The black and blue line in Fig. 3 b) represents the evaluation of this measurement data by Eq. 1 at an external compression pressure of 36.6 and 73.3 kPa. In these four measurements k_{steel} of the reference material shows a monotonous linear increase with temperature.

For the measurement uncertainties of the test bench to be evaluated, a suitable and reliable reference curve for the thermal conductivity of 1.4301 stainless steel between -10 and 50 °C is necessary. In accordance with the DIN EN 10088-3 standard [20], Stainless steel 1.4301 has a composition listed in Table 3 that is close to the AISI 304 (American Iron and Steel Institute) standard [19]. Bogaard [19] collected thermal conductivity measurements on AISI-304 from 15 references on 20 different samples within a temperature range between 200 and 1600 K. From this data collection, Bogaard [19] created one recommended curve for AISI-304 steel with a maximum deviation of $\pm 4\%$ from the measurement data. This reference curve is shown in Fig. 3 b) as a red line with the grey shaded uncertainty of $\pm 4\%$.

Comparing the red reference curve with the measurement data in black and blue in Fig. 3 reveals a maximum deviation of -2.2% which is contained within the common $\pm 4\%$ uncertainty band. Furthermore, the measurement data shows equal temperature dependency to the reference curve, and indicates no influence on the applied pressure. One important aspect for detecting changes in thermal conductivity is the reproducibility and the absence of long-time drifts in the measurement data. Comparing all blue and black curves to the black measurement at day 0 results in a maximum deviation of 0.25% with no long-time drifts within a period of at least 12 days.

2.4. Cell thermal conductivity

Once the test bench had been validated, the reference object was replaced by the cell under test described in Subsection 2.1. The test plan was the same as the validation test in Subsection 2.3, apart from increasing the duration of the rest and heating period to 7 h and reducing the heater current I_H to 1.43 A with a decline in U_H to 15.284 to 15.304 V. These adjustments were made to guarantee equal temperature gradients to the validation test with a sufficient stationary time behavior. Temperature sensors for T_{cool} and T_{hot} of the same type with the same calibration procedure were attached to the cell surface with a distance d_{sens} of 18.01 mm. Furthermore, the internal cell temperature T_{int} was recorded during every conductivity test. In total, eight conductivity tests with different external pressures were conducted.

Fig. 4 a) shows the evaluation of the measurement data by Eq. 1. Equal line colors indicate the same compression pressure starting at day 0 with 74.2 kPa as a time reference point, followed by the blue lines with a reduced pressure of 37.1 kPa after 10 days and the yellow lines with a switch back to 74.2 kPa after 24 days. Dashed lines indicate measurements with insufficient waiting time between pressure adjustments, which might be necessary due to visco-elastic relaxation processes described by Oh et al. [21]. All measurements show a rise in thermal conductivity with increasing mean temperature. Analyzing the recorded data via linear regression reveals a higher slope of

3.2 mW m⁻¹ K⁻² at 74.2 kPa than for 37.1 kPa with a value of 2.2 mW m⁻¹ K⁻². The precision measured by the individual value to the mean value of all measurements at equal preload and mean temperature shows a maximum deviation of $\pm 0.2\%$ proving an excellent measurement quality. Comparing the black and yellow lines also shows a reversible change in thermal conductivity, leading to the assumption that no plastic deformation occurred when the compression pressure changed.

2.5. Case thermal conductivity

The thermal conductivity values presented in Fig. 4 a) arise from the parallel connection of the case conductivity k_{case} and the internal part k_{int} of the battery cell. The conductivity of the case k_{case} has to be known in order to separate k_{int} from the entire cell k_{cell} . However, no suitable thermal conductivity measurements are found in literature for the aluminum alloy AW 3003 in the temperature range between -10 and 50 °C. Woodcraft introduced a method for predicting the thermal conductivity of aluminum alloys by electrical resistivity measurements that is described in [22] and used for k_{case} in this work.

For this approach, the electrical resistivity at room temperature is measured in first place. A measurement setup for this is depicted in Fig. 4 b) which applies and measures an electrical current to the smallest case surfaces and measures the potential drop along the mid length of the case with a sense contacts distance d_u of 23.18 mm. The cross-section area of the case can be computed by subtracting the outer cell area ($d_{c,h} \cdot d_{c,d}$) from the internal cell area, which is the product of ($d_{c,h} - d_{c,b} - d_{c,t}$) and ($d_{c,d} - 2 \cdot d_{c,s}$). With a cross section of the case of 196 mm² and an electrical resistance of 5.24 μΩ the case resistivity was computed to 44.19 nΩm, which closely correlates with the value listed in [23] of 43 nΩm for 3003 alloy, with the temper H18 and corresponding thermal conductivity of 155 W m⁻¹ K⁻¹.

Woodcraft's [22] approach prescribes that the electrical resistivity needs to be greater than 31 nΩm, which is fulfilled by the aluminum alloy in this work. Next, the residual resistivity due to electron-defect interactions is computed to 17.79 nΩm and a beta factor of 0.7261 is derived using Wiedemann-Franz's law, which varies with the purity of the alloy. Finally, the thermal conductivity is determined from Woodcraft's equation that is based on 29 measurements subjected to 15 different aluminum alloys. The associated results for the aluminum alloy in this work are plotted in Fig. 4 b) in addition to the constant value used by Sheng et al. [12]. To estimate the uncertainty of this method, the electrical resistance in [23] of 43 nΩm is applied in the same way with the method of Woodcraft yielding a thermal conductivity of 155.88 W m⁻¹ K⁻¹. Comparing this result with 155 W m⁻¹ K⁻¹ quantifies the uncertainty as 0.6%.

3. Theory and modeling

3.1. Unit cell conductivity

Due to the layered structure, the unit cell exhibits two faces with a through-plane $k_{\perp}$ and four faces with an in-plane thermal conductivity $k_{\parallel}$, which are defined by Eq. 2 and 3. Both parameters are affected by the layer thickness and thermal conductivity values of anode (d_A , k_A), cathode (d_C , k_C), separator (d_S , k_S), anode current collectors (d_{Cu} , k_{Cu}) and cathode current collector (d_{Al} , k_{Al}). For double-coated electrodes, all active materials and the separator have to be considered twice.

$$k_{\perp} = \frac{2 \cdot (d_A + d_S + d_C) + d_{Cu} + d_{Al}}{2 \cdot \left(\frac{d_A}{k_A} + \frac{d_S}{k_S} + \frac{d_C}{k_C} \right) + \frac{d_{Cu}}{k_{Cu}} + \frac{d_{Al}}{k_{Al}}} \quad (2)$$

$$k_{\parallel} = \frac{2 \cdot (d_A \cdot k_A + d_S \cdot k_S + d_C \cdot k_C) + d_{Cu} \cdot k_{Cu} + d_{Al} \cdot k_{Al}}{2 \cdot (d_A + d_S + d_C) + d_{Cu} + d_{Al}} \quad (3)$$

In three-dimensional thermal finite element method (FEM) models,

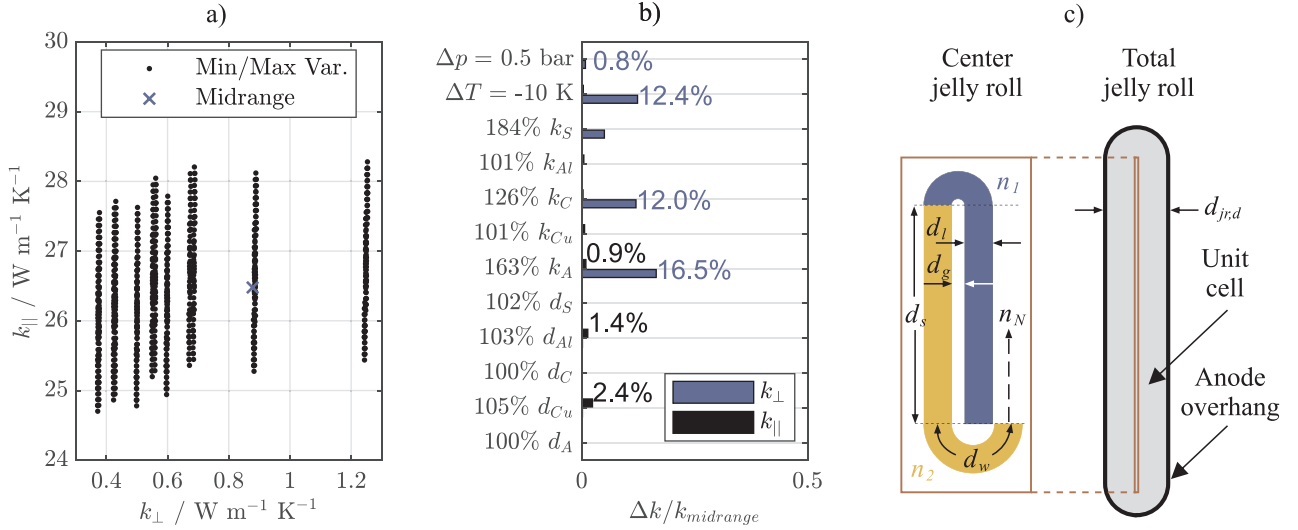


Fig. 5. a) Uncertainty range of $k_{\parallel}$ and $k_{\perp}$ for the unit cell taking k_{Max} and k_{Min} from Table 4 and uncertainties in layer thicknesses. b) Sensitivity of $k_{\perp}$ and $k_{\parallel}$ around mid-range values from a) with changes in pressure Δp , temperature ΔT , thermal conductivity values of unit cell layers (k_S , k_{Al} , k_C , k_{Cu} , k_A) and layer thicknesses (d_S , d_{Al} , d_C , d_{Cu} , d_A). c) Geometric definition for estimating the thickness of the jelly roll gap d_g .

$k_{\parallel}$ and $k_{\perp}$ are often parameterized by measurements on unit cell layers [24] that have no dependency on pressure or temperature. To estimate the uncertainties of this method, Table 4 lists minimum k_{Min} and maximum k_{Max} values measured on layer level. Further uncertainties in Eq. 2 and 3 arise from measurement errors of the layer thicknesses that are quantified to $\pm 0.5 \mu\text{m}$ for common micrometers.

To illustrate the impact of these uncertainties, Eq. 2 and 3 are applied 2^{10} times using all possible combinations of minimum and maximum values for the thermal conductivity or layer thickness. Black dots in Fig. 5 a) show the results of this min/max variation with a range from 0.371 to 1.254 $\text{W m}^{-1} \text{K}^{-1}$ for $k_{\perp}$ and 24.7 to 28.3 $\text{W m}^{-1} \text{K}^{-1}$ for $k_{\parallel}$. Considering the mid-range of k_{Min} and k_{Max} as the most likely scenario leads to values for $k_{\perp}$ and $k_{\parallel}$ of 0.878 and 26.5 $\text{W m}^{-1} \text{K}^{-1}$ highlighted by the blue cross in Fig. 5 a). Dividing the mid-range result by the largest deviation from the min/max variation computes to an uncertainty for $k_{\parallel}$ of 6% and for $k_{\perp}$ of 58%.

Fig. 5 b) illustrates the sensitivity of $k_{\parallel}$ and $k_{\perp}$ by varying only one parameter around the mid-range value. For example, the maximum value for k_C amounts to 126% of its mid-range of 0.725 $\text{W m}^{-1} \text{K}^{-1}$

affecting $k_{\perp}$ by 12%. Uncertainties for the thicknesses ranging from 100 to 105% are calculated with a measurement uncertainty of $0.5 \mu\text{m}$ regarding the thickness values in Table 2. According to Fig. 5 b), $k_{\perp}$ is sensitive to the thermal conductivity values of the active materials (k_A , k_C) and the separator k_S , while $k_{\parallel}$ is mostly affected by the thickness values of the current collectors (d_{Al} , d_{Cu}).

Table 4 lists the relative pressure dependency $(dk/dp)/k_{2\text{bar}}$ for changes in thermal conductivity and the Young's modulus E for changing layer thicknesses ε to estimate the pressure dependency of $k_{\parallel}$ and $k_{\perp}$. When pressure is increased by 0.5 bar, the active materials (d_A , d_C) and the separator thicknesses d_S decline by -0.1 to -0.3%. Taking the changes in layer thickness in combination with $(dk/dp)/k_{2\text{bar}}$ for a pressure increase of 0.5 bar alters the mid-range value in Fig. 5 a) by 0.2% for $k_{\parallel}$ and 0.8% for $k_{\perp}$.

The temperature dependency of the layer thermal conductivity $(dk/dT)/k_{20}$ and the linear thermal expansion coefficient α are listed in Table 4. For $(dk/dT)/k_{20}$ no measurement data on electrolyte soaked NMC, separator or graphite could be found in literature. Therefore, missing temperature dependencies are taken from Werner et al. [8],

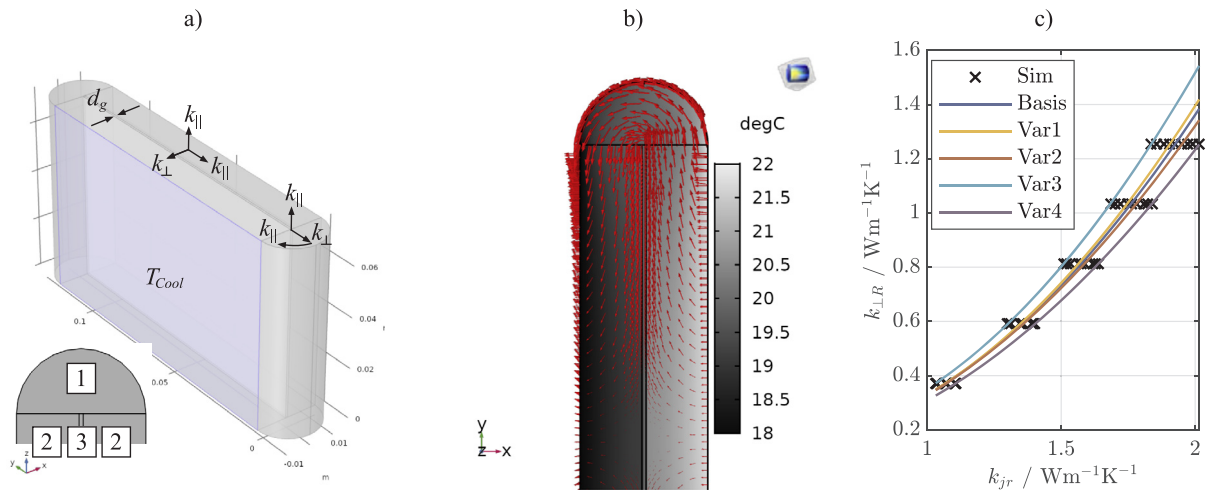


Fig. 6. a) Geometry of thermal FEM model consisting of the volumes windings (1), stack (2) and jelly roll gap (3) used for the functional relation between $k_{\parallel}$ and $k_{\perp}$. b) Heat flow (red arrows) and temperature field inside the jelly roll with $d_g = 0.8 \text{ mm}$ and a temperature gradient of $4 \text{ }^{\circ}\text{C}$ between the jelly roll case contact areas. c) Functional relation between $k_{\parallel}$ and $k_{\perp}$ for the basis scenario and variation in d_g and $k_{\parallel}$ specified in Table 5. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

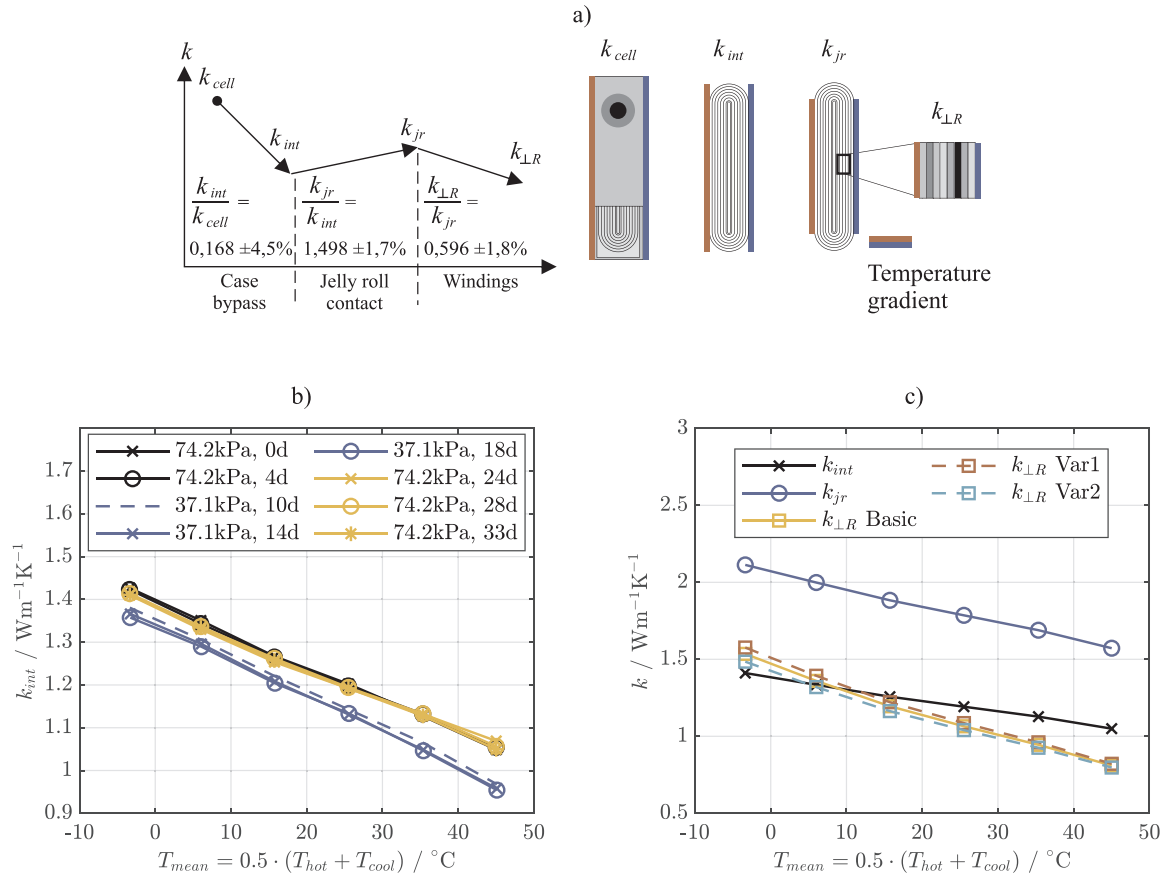


Fig. 7. a) Overview of transformation process from k_{cell} to $k_{\perp R}$ with transformation factors and their uncertainties for the data presented in c) at 25.5 °C. b) Internal cell conductivity k_{int} after the case bypass from Fig. 4 a) has been compensated for. c) Exemplary transformation from k_{int} to $k_{\perp R}$ for the measurement started at day 33.

which are measured on dry samples and calculated into electrolyte-soaked values. To estimate the influence of a -10 °C temperature drop on $k_{||}$ and $k_{\perp}$, the layer thicknesses are multiplied by $(1 + \alpha \cdot (-10 ^\circ \text{C}))$ and the mid-range of k_{max} and k_{min} is multiplied by $(1 + (dk/dT)/k_{20} \cdot (-10 ^\circ \text{C}))$. When these adjustments are combined with Eq. 2 and 3, this leads to a change in $k_{||}$ of 0.4% and $k_{\perp}$ by 12.4%. This postulates a strong temperature dependency for $k_{\perp}$ of $-1.24\% \text{K}^{-1}$.

3.2. Jelly roll gap

During cell production, the electrode-separator composite is rolled around a winding core [35]. Due to this winding process, an argon-filled gap in the middle of the jelly roll may remain, which is illustrated in Fig. 5 c). Bandhauer et al. [14] assumed the thickness of this gap d_g to be 0.5 mm for a prismatic jelly roll with a unit cell thickness of approximately 0.28 mm. For external heat flows passing the jelly roll between its largest case contact faces, this jelly roll gap might constitute

Table 4

Parameters for calculating $k_{||}$ and $k_{\perp}$ according to Eq. 2 and 3 with measurements on unit cell layers. The minimum k_{Min} and maximum k_{Max} reported thermal layer conductivity in literature, the pressure dependency of the thermal conductivity $(dk/dp)/k_{2bar}$, the Young's modulus E and corresponding strain ϵ at 0.5 bar, the temperature dependency $(dk/dT)/k_{20}$ of the cell layers and the linear thermal expansion coefficient α are used to evaluate uncertainties and the temperature or pressure dependency of $k_{||}$ and $k_{\perp}$.

Unit cell layer	$k_{Min} / \text{W m}^{-1} \text{K}^{-1}$	$k_{Max} / \text{W m}^{-1} \text{K}^{-1}$	$(dk/dp)/k_{2bar} / \% \text{ bar}^{-1}$	E / MPa	$\epsilon(0.5 \text{ bar}) / \%$	$(dk/dT)/k_{20} / \% \text{ K}^{-1}$	$\alpha / 10^{-6} \text{ K}^{-1}$
NMC ^a	0.54 [25] ^b	0.91 [28] ^b	0.5 [28]	40 [30] ^g	-0.1	n/a ^d	12 - 13 [32]
Al (pure)	235 [26]	240 [26]	^c	$70 \cdot 10^3$ [31]	0.0	0.0 [26]	23.1 [31]
Separator ^a	0.10 [25] ^b	1.13 [29]	-0.6 ^h [25]	20 [30] ^g	-0.3	n/a ^e	44 [33]
Graphite ^a	0.36 [27] ^b	1.58 [25] ^b	3.95 ⁱ [25,28]	33 [30] ^g	-0.2	n/a ^f	55 [34]
Cu (pure)	394 [26]	404 [26]	^c	$120 \cdot 10^3$ [31]	0.0	0.0 [26]	16.5 [31]

^a Soaked with electrolyte.

^b Measured around room temperature.

^c No dependency is assumed.

^d For LCO cathode derived from dry measurements $-1.7\% \text{K}^{-1}$ [8].

^e Derived from dry measurements $-0.0\% \text{K}^{-1}$ [8].

^f Derived from dry measurements $-1.0\% \text{K}^{-1}$ [8].

^g For 0.2 MPa.

^h Mid-range of $-3.4 - 2.2 \% \text{ bar}^{-1}$.

ⁱ Mid-range of $0.9 - 7.0 \% \text{ bar}^{-1}$.

Table 5

Fitting parameters for Eq. 8 shown in Fig. 6 c) for the functional relation between k_{jr} and k_{LR} .

	Basis	Var1	Var2	Var3	Var4
d_g/mm	0.8	1.0	0.6	1.0	0.6
$k_{ }/\text{W m}^{-1} \text{K}^{-1}$	26.5	26.5	26.5	24.7	28.3
$a/\text{W}^{-1} \text{m K}$	0.4474	0.4818	0.4038	0.5294	0.3739
b	-0.3091	-0.3802	-0.2173	-0.4222	-0.1972
$c/\text{W m}^{-1} \text{K}^{-1}$	0.1903	0.2283	0.1406	0.2442	0.1319

a significant heat barrier, because argon exhibits a low thermal conductivity of $0.0179 \text{ W m}^{-1} \text{K}^{-1}$ [31]. As external heat flows are applied for the test procedure in Section 2, the influence of the jelly roll gap requires further research. The most important parameter for this study is the thickness of the jelly roll gap d_g , which is estimated using the relationship between the total electrode length d_{el} , the thickness of the electrode separator composite d_l and the thickness of the flat-wound jelly roll $d_{jr,d}$.

To determine this relationship, the length of the straight jelly roll sections d_s and the half windings d_w is added N times until the value of d_{el} is reached. For simplicity, in Eq. 4 every winding process comprises a straight and a winding section. In contrast to d_s , the length of the half winding section depends on the winding number n and the gap thickness d_g as stated by Eq. 5. Inserting Eq. 5 into 4 and applying the Gaussian arithmetic series allows Eq. 6 to be derived. The total thickness of the wound electrode d_{el} can be calculated from Eq. 7 with the knowledge of N .

$$d_{el} = \sum_{n=1}^N (d_w(n) + d_s) \quad (4)$$

$$d_w(n) = 0.5 \cdot \pi \cdot (n \cdot d_l + d_g) \quad (5)$$

$$d_{el} = N \cdot (d_s + 0.5 \cdot \pi \cdot d_g) + 0.5 \cdot \pi \cdot d_l \cdot \frac{N \cdot (N + 1)}{2} \quad (6)$$

$$d_{el} = N \cdot d_l + d_g \quad (7)$$

Due to the internal sensors, the capacity of the cell under test decreased by 7%, which is translated to a 7% shorter unit cell length $d_{el,uc}$ of 10.137 m compared to the cathode length in Table 2. Solving Eq. 6 with this value for $d_{el,uc}$, an assumed $d_{g,uc}$ of 1 mm for the internal sensors, d_s of 117 mm according to Fig. 1 and a value for $d_{l,uc}$ of 315.2 μm results in 74 unit cell layers N_{uc} . According to Eq. 7, the corresponding thickness $d_{el,uc}$ is equal to 24.32 mm.

Furthermore, the anode overhang discussed in Subsection 2.1 is considered in order to estimate the total thickness of the jelly roll. This overhang has a total length $d_{el,ao}$ of 600 mm and a thickness $d_{l,ao}$ of 112.7 μm , with a single layer for the current collector d_{Cu} and anode coating d_A and two separator layers d_s . Assuming the overhang appears only on the outer windings, $d_{g,ao}$ is equal to $d_{el,uc}$ of the unit cell winding. Solving Eq. 6 for the anode overhang with the same value for d_s as for the unit cell results in four anode overhang layers N_{ao} . The overall thickness of the jelly roll $d_{jr,d}$ is 24.8 mm, which is a plausible value, as in its dry state the jelly roll has to fit into the 25 mm of free space between the case walls.

Due to solid electrolyte interface formation and particle rearrangement, the thickness of the unit cell in its dry state increases during the formation process [36]. This irreversible and the total dilation is calculated by the work of Sauerteig et al. [37] with the knowledge of the cathode mass loading and the porosity of the anode. Applying these equations with a cathode mass loading of 21.1 mg cm^{-2} results in an irreversible dilation of 2.51 μm and a total dilation of 6.55 μm . Adding 2.51 μm to $d_{l,uc}$ in its dry state will increase $d_{jr,d}$ to 25.0 mm after the formation process at zero state of charge. The remaining dilation of 4.04 μm until total dilation is reached constitutes a 1.3% increase in $d_{l,uc}$

compared to the thickness after formation. This increase from zero to full state of charge matches the range of Bauer et al. [36] of 1–2% for a NMC graphite unit cell. Taking the 2% as an extreme scenario will rise $d_{jr,d}$ to 25.4 mm. Assuming the free space between the case walls stays constant, d_g changes between 0.6 and 1.0 mm.

3.3. From unit cell to jelly roll conductivity

As already discussed by Werner et al. [8], the main thermal contact between the electrodes inside a prismatic cell with their environment is at the two largest surfaces of the case. Considering a planar stack without windings and a temperature gradient between the largest cell faces, each heat flow is influenced only by $k_{\perp}$. In the case of a flat-wound electrode geometry, the overall jelly roll conductivity is also affected by $k_{||}$ due to heat flows through the windings.

In this study, the approach by Werner et al. [8] was used under the boundary condition of this work and the consideration of the jelly roll gap. Furthermore, instead of analytic calculation, a three-dimensional FEM model was created in COMSOL Multiphysics 5.4 to evaluate the correlation between $k_{\perp}$ and k_{jr} . The geometry of this model is depicted in Fig. 6 a) with a total height in the z-direction equal to the height of the cathode listed in Table 2. The length in the y-direction of the case contact area indicated in purple is set to d_s , while the overall thickness in the x-direction amounts to $d_{jr,d}$. The total jelly roll is subdivided into two winding volumes (tag 1), two electrode stack volumes (tag 2) and one volume for the jelly roll gap (tag 3) defined by d_g . The thermal conductivity of the winding and stack volumes are anisotropic with a cylindrical coordinate system for the winding section and a Cartesian for the stacked volumes. For any simulation, the thermal conductivity is assumed to be independent of temperature. All boundaries are thermally isolated except for the two case contact areas with an isothermal boundary of T_{cool} and T_{hot} of 18 °C and 22 °C.

Fig. 6 b) shows the simulation results for a jelly roll gap of 0.8 mm and a through-plane and in-plane conductivity of 0.8125 and 26.5 $\text{W m}^{-1} \text{K}^{-1}$. The surface integral over the case contact areas results in a total heat of 2.107 W, leading to an effective thermal conductivity between the largest stack surfaces of 1.575 $\text{W m}^{-1} \text{K}^{-1}$. The red vector field illustrates the direction of the heat flow with the arrow length on a logarithmic scale. According to this simulation, the major heat flow is conducted through the winding volumes. The surface integral of the heat flux entering the gap volume gives rise to a heat of 0.1975 W which accounts for 9.4% of the total heat flux. This fraction increases to 28.7% if no gap volume is present.

The primary aim of the simulation is to establish a correlation between the through-plane conductivity of the unit cell with contact resistances k_{LR} and the jelly roll k_{jr} that is influenced by d_g and $k_{||}$. Both parameters can only be determined by their ranges of 0.6 to 1.0 mm and 24.7 to 28.3 $\text{W m}^{-1} \text{K}^{-1}$. Therefore, a parametric sweep was performed with the mentioned ranges and four equidistant values for k_{LR} between 0.3710 and 1.2188 $\text{W m}^{-1} \text{K}^{-1}$. The simulation results are displayed in Fig. 6 c) as black crosses. For constant values for d_g and $k_{||}$ the correlation between k_{LR} and k_{jr} reveals a quadratic relationship. Five quadratic least square fits are depicted in Fig. 6 c) with the blue line demonstrating the most likely basis scenario, with mid-range values for d_g of 0.8 mm and $k_{||}$ of 26.5 $\text{W m}^{-1} \text{K}^{-1}$. The yellow (Var1) and red (Var2) lines indicate only a variation in the parameter d_g marking the uncertainties in the jelly roll gap. The two remaining variations (Var3 and 4) in purple and cyan constitute the upper and lower bound by varying d_g and $k_{||}$. All parameters for Fig. 6 c) and Eq. 8 are listed in Table 5.

$$k_{LR} = a \cdot k_{jr}^2 + b \cdot k_{jr} + c \quad (8)$$

3.4. From cell to jelly roll conductivity

Fig. 7 illustrates the overall evaluation process, starting with the measured cell conductivity k_{cell} and ending at the quantity of interest k_{LR} . The case bypass has to be first compensated for leading to the internal conductivity k_{int} , which consists of the gas volume inside the cell and the jelly roll. The thermal conductivity k_{jr} of the jelly roll is derived from the knowledge of the contact area between the jelly roll and the cell case. Subsection 3.3 addressed the correlation between k_{jr} and k_{LR} . To complete the evaluation process from k_{cell} to k_{LR} , the relation between k_{cell} and k_{jr} is explained in this subsection.

To evaluate the internal cell conductivity k_{int} , Eq. A.3 has to be resolved to k_{int} , which is shown by Eq. 9. The conductivity value for the case k_{case} is extracted from Fig. 4 b) with the same mean temperature as the cell measurement.

$$k_{int} = \frac{A_{cell}}{A_{cell} - A_{case}} \cdot k_{cell} - \frac{A_{case}}{A_{cell} - A_{case}} \cdot k_{case} \quad (9)$$

Applying Eq. 9 to k_{cell} in Fig. 4 will result in the internal conductivity k_{int} in Fig. 7 b). Due to the high value of k_{case} , k_{int} is approximately six times smaller than k_{cell} . Furthermore, the dependency on the mean temperature T_{mean} alters its sign from a rising to a falling characteristic with increasing temperature.

The correlation between k_{int} and k_{jr} in Eq. 10 is defined by the ratio of the internal surface of the case A_{int} and the contact area between the case and the jelly roll A_{jr} . A_{int} is determined by Eq. A.2 and A_{jr} is the product of the cathode height in Table 2 and the jelly roll contact length d_s .

$$k_{jr} = \frac{A_{int}}{A_{jr}} \cdot k_{int} \quad (10)$$

An example of the overall chain of transformation from k_{int} to k_{LR} is illustrated in Fig. 7 c) for the cell measurement that started at day 33, with a compression pressure of 74.2 kPa. At 25.5 °C, k_{int} amounts to 1.192 W m⁻¹ K⁻¹ after the case is compensated for. Moreover, k_{int} is converted to k_{jr} using Equation 10, calculating to a conductivity value of 1.786 W m⁻¹ K⁻¹. By applying Eq. 8, k_{LR} amounts to 1.065 W m⁻¹ K⁻¹ for the basis scenario with a range of 0.972 (Var2) and 1.178 W m⁻¹ K⁻¹ (Var1).

To analyze the uncertainty for k_{LR} , the root square sum of each uncertainty in the chain of transformation shown in Fig. 7 a) is calculated. This approach is in accordance with the uncertainty analysis of Sheng et al. [12]. The maximum uncertainty of 2.2% for k_{cell} was determined in Subsection 2.3. The case bypass introduces an uncertainty of 4.5% including errors in the approach by Woodcraft and the electrical resistivity measurement. Assuming an absolute error for d_s of 2 mm quantifies the uncertainty of the jelly roll contact to be 1.7%. Finally, 1.8% uncertainty exists, which arises from the jelly roll gap variation Var1 and Var2. Building the root square sum of all uncertainties results in a total value of 5.6% for k_{LR} .

4. Results and discussion

In Fig. 8, the conductivity k_{LR} for external compression pressures of 37.1 a) and 74.2 kPa b) is analyzed according to the procedure presented in Section 3. Cross markers show the transformed measurement data from Fig. 4 a) and their best linear fits are highlighted in solid lines. These three lines represent the most likely scenario (Bias, black) and the lower (Var2) and upper (Var1) bound regarding the uncertainty for the jelly roll gap d_g . The gray-shaded area fills the span between Var3 and Var4 including uncertainties in d_g and $k_{||}$. No linear fit is drawn for Var3 and Var4 in Fig. 8, because $k_{||}$ is assumed to not change significantly with differences in temperature or pressure, which was discussed in Section 3.

Values for k_{LR} taken at 20 °C for the basis scenario represented by the black lines ranging from 1.047 to 1.171 W m⁻¹ K⁻¹ correlate closely

with the range of 0.371 and 1.254 W m⁻¹ K⁻¹ calculated from the unit cell layer measurements in Fig. 5. Higher values up to 1.6 W m⁻¹ K⁻¹ in Fig. 8 can be explained by the fact that most values in Table 4 are measured at around room temperature.

The following subsections discuss how k_{LR} change with changes in external pressure and mean temperature of the cell. Treating k_{LR} as constant, will cause errors in the determination of temperature gradients inside the cell. According to the analysis in Appendix B, an overestimation of k_{LR} of +10% will translate to an error in temperature gradient of -9.1%.

4.1. Dependency on external pressure

An increase of about (11.9 ± 0.3)% at 20 °C is observed for k_{LR} when the external compression pressure is changed from 37.1 to 74.2 kPa for the basic scenario and the variations Var1 and Var2. This enhancement can be the result of reduced thermal contact resistances R_{tc} , a smaller jelly roll gap d_g or an increased thermal conductivity of the material $k_{\perp}$ at 74.2 kPa. An increase of 37.1 kPa at the largest cell surfaces converts to a rise in pressure at the jelly roll to case contact of approximately 0.5 bar, if no significant interaction of the case is assumed. According to Fig. 5 b), this enlargement in through-plane pressure affects $k_{\perp}$ by an increase of only 0.8%. Referring to the strain values in Table 4, the cell thickness will change less than 0.3% at this pressure increase. This precludes any dominant effect arising from changes in d_g or d_{sens} . Therefore, only changing thermal contact resistances R_{tc} appears as the main cause for the observed changes in k_{LR} .

Eq. 11 [38] calculates the effect on the overall thermal conductivity k_{LR} by the sum of thermal contact resistances R_{tc} between the layers and the case-separator interface. Hence, R_{tc} is determined by Eq. 12 with the knowledge of k_{LR} and $k_{\perp}$. The maximum value for R_{tc} at 37.1 kPa can be estimated by the highest value of $k_{\perp}$ of 1.254 W m⁻¹ K⁻¹ in Fig. 5 and $k_{LR}(20\text{ °C}, 37.1\text{ kPa})$ in Fig. 8 of 1.047 W m⁻¹ K⁻¹. According to this estimation, R_{tc} is lower than 553 μm²KW⁻¹. This range correlates with the results found by Ponnappan et al. [9], who measured values between 300 and 400 μm²KW⁻¹ and the results of Ye et al. [38] of 150 μm²KW⁻¹.

$$k_{LR}(T, p) = \frac{d_{l,uc}}{\frac{d_{l,uc}}{k_{\perp}(T)} + R_{tc}(p)} \quad (11)$$

$$R_{tc}(p) = \frac{d_{l,uc}}{k_{LR}(T, p)} - \frac{d_{l,uc}}{k_{\perp}(T)} \quad (12)$$

$$R_{tc}(37.1\text{ kPa}) - R_{tc}(74.2\text{ kPa}) = \frac{d_{l,uc}}{k_{LR}(T, 37.1\text{ kPa})} - \frac{d_{l,uc}}{k_{LR}(T, 74.2\text{ kPa})} \quad (13)$$

The absolute value for R_{tc} can not be determined with the measurement results in Fig. 8. Nevertheless, by the assumption of a pressure independent thermal conductivity $k_{\perp}(T)$ the difference between $R_{tc}(37.1\text{ kPa})$ and $R_{tc}(74.2\text{ kPa})$ can be derived from Eq. 13. For the Basic scenario at 20 °C this difference amounts to 32.2 μm²KW⁻¹. In the results of Ponnappan et al. [9], R_{tc} reached saturation at 300 μm²KW⁻¹ for pressures higher than 50 psi (3.4 bar). This gives rise to the assumption of a further enhancement of k_{LR} of the cell investigated here, if the external compression pressure is further increased.

4.2. Dependency on temperature

Dividing the slopes of the linear regression of the base cases in Fig. 8 by the thermal conductivity value at 20 °C results in a temperature dependence of -1.49%K⁻¹ at 37.1 kPa and -1.27%K⁻¹ at 74.2 kPa. These values correspond with the calculated values in Fig. 5 a) of -1.24%K⁻¹ and correlate closely with the measurement results by Werner et al. [8] shown in Table 1. Nevertheless, these values disagree with the observations made by Bazinski et al. [11] and Sheng et al. [12].

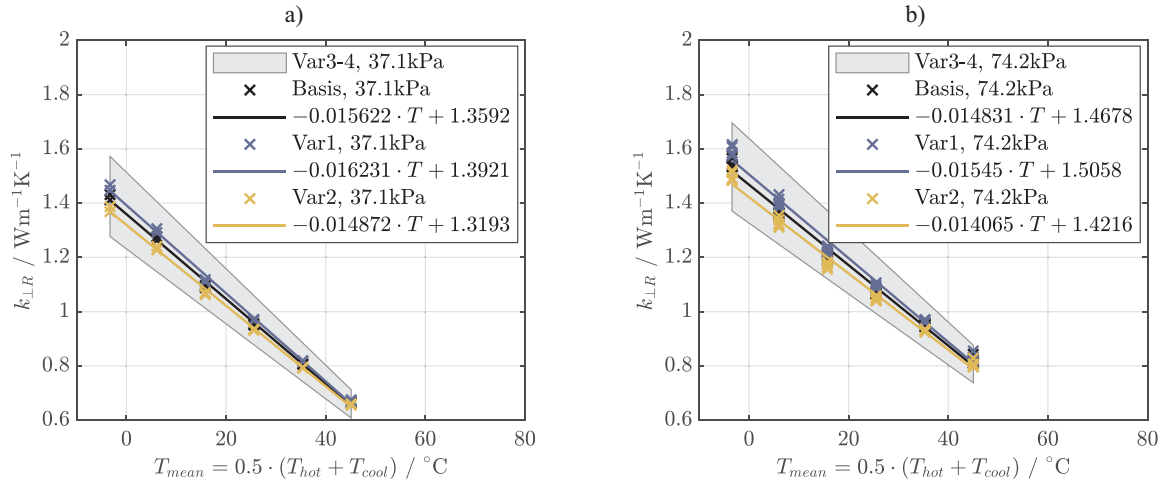


Fig. 8. Through-plane thermal conductivity of the unit cell k_{LR} at an external compression pressure of 37.1 a) and 74.2 kPa b). For deriving the temperature dependency, linear regression parameters are given for the basis scenario and the parameter variation Var1 and Var2, which considers different jelly roll gap thicknesses d_g . Var3 and Var4 include uncertainties in $k_{||}$ and d_g marking the extreme upper and lower bound.

Several factors affect the temperature dependency of k_{LR} [13] [3], which also explain the different findings in literature:

1. Material composition (e.g. binder, electrolyte, active materials, carbon black, separator coatings).
2. Electrode structure (e.g. porosity, particle size distribution and shape, particle-particle interface).
3. Thickness ratio of inactive to active material (d_A/d_{Cu} d_C/d_{Al}).
4. Layer-to-layer contact resistance R_{ct} .

Most investigations listed in Table 1 reveal less detailed information on material composition or electrode structure. However, the LFP cathode system is characterized by a low temperature dependency resulting from lower particle sizes around 2 μm [39], compared to NMC with sizes around 10 μm [40]. Maleki et al. [3] measured the thermal conductivity of negative electrode material without electrolyte for particle sizes ranging from 6 to 75 μm . Compared to the base sample with a temperature dependency of $-0.01\%K^{-1}$ and a particle size of 6 μm , the temperature dependency for a sample with a particle size of 75 μm decreases to $-0.38\%K^{-1}$.

PVDF binder, applied in most commercial active materials, might be one of the main causes for the negative temperature dependency. In the work of dos Santos et al. [41], the thermal conductivity of PVDF decreased from 0.26 to 0.18 $W m^{-1} K^{-1}$ for a temperature increase from 20 to 100 $^{\circ}C$. This corresponds to a temperature dependence of $-0.56\%K^{-1}$. At temperatures higher than 100 $^{\circ}C$, the thermal conductivity of PVDF starts rising rapidly. Based on $(dk/dT)/k_{20}$ of PVDF, it can be assumed, that k_{LR} in Fig. 8 will keep decreasing until 100 $^{\circ}C$. Furthermore, varying weight portion of PVDF could explain different findings in Table 1.

As discussed by Loges et al. [13], the thickness ratio of electrodes influences the overall thermal conductivity. This also applies for the temperature dependency, which gets closer to zero if the current collectors become more dominant. Therefore, the low values of d_A/d_{Cu} and d_C/d_{Al} given by Sheng [12] listed in Table 1 may be the reason for the positive temperature dependency observed in the measurements for the prismatic LFP hardcase cell.

Thermal contact resistance between the cell layers can change the temperature dependency of k_{LR} . Ponnappan et al. [9] and Gaitonde et al. [42] observed no significant change of R_{tc} over temperature. A dominant temperature independent R_{tc} could compensate any temperature dependence of k_{LR} close to zero. Nevertheless, due to thermal swelling, thermal contact resistances may also influence the slope of the linear curve by changing the layer-to-layer pressure.

5. Conclusion

This work investigates the through-plane thermal conductivity inside a prismatic NMC lithium-ion cell at different temperatures and external pressures. For detecting changes in thermal conductivity, high precision and accuracy of the measurement procedure is inevitable. Therefore, a test bench is designed in compliance with the guarded heater principle and validated extensively with a standard reference material. The validation proves excellent measurement quality for the test bench, which is adaptable to different prismatic formats and pouch cells.

The hardcase of prismatic cells constitutes the major thermal interface between the inner jelly roll and the cell surrounding. The thermal conductivity of the case is thus an important quantity for this work. Therefore, a nondestructive measurement technique was introduced that determines the thermal conductivity of the cell case by means of electrical resistivity measurements. This method was first applied in the field of battery research and is useful for any thermal investigation with prismatic cells.

Most prismatic hardcase cells have to be installed with an external pressure to counter swelling of the jelly roll. The manufacturer of the PHEV2 cell in this work specified this pressure between 37.1 and 74.2 kPa. Increasing the pressure within this range rises the thermal through-plane conductivity by 11.9%. Thermal contact resistances between the cell layers of 32.2 $\mu m^2 KW^{-1}$ explained this thermo-mechanical interaction. Future work will look into measurements with more than two pressure values to gain more information on the course of thermal conductivity.

The thermal conductivity of the investigated cell decreases linearly with increasing temperature. Compared to the absolute value at 20 $^{\circ}C$, this decrease amounts to more than 1% per K. Taking this temperature dependency into account may improve thermal modeling at extreme temperatures, which is especially the case in the area of abuse simulations. Therefore, future studies on the progression of the thermal conductivity above 40 $^{\circ}C$ are recommended.

CRedit authorship contribution statement

Marco Steinhardt: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing - original draft. **Elisabeth Irene Gillich:** Methodology, Validation, Writing - review & editing, Funding acquisition. **Maximilian Stiegler:** Validation, Software, Writing - review & editing.

Andreas Jossen: Writing - review & editing, Funding acquisition.

Acknowledgements

This work was supported financially by the German Federal Ministry of Education and Research (BMBF) [03XP0027G MiBZ] and Federal Ministry for Economic Affairs and Energy (BMWi) [03ET6153C iMoBatt]. The authors thank Korbinian Schmidt from the Institute for Electrical Energy Storage Technology at the Technical University of Munich for his practical support.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Selection of reference material

The reference material in this study was selected in accordance with four selection criteria stated by Hust et al. [17]:

1. The material should be homogeneous and isotropic.
2. Thermal conductivity should not vary with time.
3. The material should be available, machinable and inexpensive.
4. The material should have similar thermal conductivity as the material to be measured.

According to Maglić et al. [43], possible candidates meeting the first three criteria are copper ($400 \text{ W m}^{-1} \text{ K}^{-1}$), tungsten ($173 \text{ W m}^{-1} \text{ K}^{-1}$), austenitic stainless steel ($15 \text{ W m}^{-1} \text{ K}^{-1}$) and Pyrex glass ($1 \text{ W m}^{-1} \text{ K}^{-1}$). For the fourth criterion to be met, the thermal conductivity of the cell k_{cell} has to be estimated. In this investigation, the heat flow is applied between the largest cell surfaces passing through the case and the jelly roll. The thermal conductivity of this parallel connection k_{cell} can be calculated by Eq. A.1, which is a weighted average of k_{case} and the inner part of the cell k_{int} mainly influenced by the jelly roll. The weighting is defined by the cross-section areas for the outer dimension of the full cell A_{cell} and the internal area A_{int} , which are defined by Eq. A.2 and A.3.

$$k_{cell} = \frac{k_{int} \cdot A_{int} + k_{case} \cdot (A_{cell} - A_{int})}{A_{cell}} \quad (\text{A.1})$$

$$A_{int} = (d_{c,h} - d_{c,t} - d_{c,b}) \cdot (d_{c,l} - 2 \cdot d_{c,s}) \quad (\text{A.2})$$

$$A_{cell} = d_{c,h} \cdot d_{c,l} \quad (\text{A.3})$$

Suitable values for k_{int} and k_{case} are necessary for estimating k_{cell} . Werner et al. [8] measured values between 1.0 and $1.5 \text{ W m}^{-1} \text{ K}^{-1}$ for the whole jelly roll. Taking the mid-range of $1.25 \text{ W m}^{-1} \text{ K}^{-1}$ for k_{int} and the mid-range of AW-3003 of $175 \text{ W m}^{-1} \text{ K}^{-1}$ results in $7.99 \text{ W m}^{-1} \text{ K}^{-1}$ for k_{cell} . This result narrows the choice of suitable reference material to either austenitic stainless steel or Pyrex glass. As stainless steel was already used by Bazinski [11] in the field of battery research, this material was applied for quantifying the measurement error in this study.

Appendix B. Effects on temperature gradient

Werner et al. [8] described the one-dimensional temperature distribution $T(x)$ from the cell core ($x = 0$) to the cell surface ($x = X$) by Eq. B.1 at steady-state condition ($dT(x)/dt = 0$). For this equation, the volumetric heat generation $\dot{q}$ is assumed constant over time and space and an adiabatic boundary condition is applied in the cell core. At the cell surface, a convective boundary with a heat transfer coefficient h and a cooling fluid temperature T_{fluid} is chosen. Eq. B.2 defines the incorrect temperature distribution $T_e(x)$ by multiplying k_{LR} with an relative error in thermal conductivity e_k .

$$T(x) = T_{fluid} + \frac{\dot{q} \cdot X^2}{2 \cdot k_{LR}} \left(1 - \frac{x^2}{X^2} + \frac{2 \cdot k_{LR}}{h \cdot X}\right) \quad (\text{B.1})$$

$$T_e(x) = T_{fluid} + \frac{\dot{q} \cdot X^2}{2 \cdot k_{LR} \cdot (1 + e_k)} \left(1 - \frac{x^2}{X^2} + \frac{2 \cdot k_{LR} \cdot (1 + e_k)}{h \cdot X}\right) \quad (\text{B.2})$$

The quadratic Eq. B.1 has an absolute maximum at $x = 0$. Therefore, the maximum temperature gradient inside the cell exists between the cell core $T(x = 0)$ and the cell surface $T(x = X)$. Uncertainties in k_{LR} have an impact on this temperature difference. Eq. B.3 defines this impact by the relative error in the maximum temperature gradient inside the cell $e_{\Delta T}$. Combining the Eqs. B.1 to B.3, will result in the relation between $e_{\Delta T}$ and e_k in Eq. B.4.

$$\frac{T_e(x = 0) - T_e(x = X)}{T(x = 0) - T(x = X)} = 1 + e_{\Delta T} \quad (\text{B.3})$$

$$e_{\Delta T} = \frac{1}{1 + e_k} - 1 \quad (\text{B.4})$$

According to Eq. B.4, $e_{\Delta T}$ is only affected by e_k at steady-state conditions. Overestimating k_{LR} by $+10\%$ will therefore result in underestimating the temperature gradient by $e_{\Delta T} = -9.1\%$.

References

- [1] Y. Liu, Y. Zhu, Y. Cui, Challenges and opportunities towards fast-charging battery materials, *Nat. Energy* 4 (7) (2019) 540–550, <https://doi.org/10.1038/s41560-019-0405-3>.
- [2] S. Ahmed, I. Bloom, A.N. Jansen, T. Tanim, E.J. Dufek, A. Pesaran, A. Burnham, R.B. Carlson, F. Dias, K. Hardy, M. Keyser, C. Kreuzer, A. Markel, A. Meintz, C. Michelbacher, M. Mohanpurkar, P.A. Nelson, D.C. Robertson, D. Scofield, M. Shirk, T. Stephens, R. Vijayagopal, J. Zhang, Enabling fast charging – a battery

- technology gap assessment, *J Power Sources* 367 (2017) 250–262, <https://doi.org/10.1016/j.jpowsour.2017.06.055>.
- [3] H. Maleki, J. Selman, R. Dinwiddie, H. Wang, High thermal conductivity negative electrode material for lithium-ion batteries, *J Power Sources* 94 (1) (2001) 26–35, [https://doi.org/10.1016/S0378-7753\(00\)00661-3](https://doi.org/10.1016/S0378-7753(00)00661-3).
- [4] R. Spotnitz, J. Franklin, Abuse behavior of high-power, lithium-ion cells, *J Power Sources* 113 (1) (2003) 81–100, [https://doi.org/10.1016/S0378-7753\(02\)00488-3](https://doi.org/10.1016/S0378-7753(02)00488-3).
- [5] T. Waldmann, G. Bisle, B.-I. Hogg, S. Stumpp, M.A. Danzer, M. Kasper, P. Axmann, M. Wohlfahrt-Mehrens, Influence of cell design on temperatures and temperature gradients in lithium-ion cells: an in operando study, *J Electrochem Soc* 162 (6) (2015) A921–A927, <https://doi.org/10.1149/2.0561506jes>.
- [6] T. Waldmann, M. Wohlfahrt-Mehrens, In-operando measurement of temperature gradients in cylindrical lithium-ion cells during high-current discharge, *ECS Electrochemistry Letters* 4 (1) (2015) A1–A3, <https://doi.org/10.1149/2.0031501eel>.
- [7] M. Klett, R. Eriksson, J. Groot, P. Svens, K. Ciosek Högstöm, R.W. Lindström, H. Berg, T. Gustafson, G. Lindbergh, K. Edström, Non-uniform aging of cycled commercial lifepo4/graphite cylindrical cells revealed by post-mortem analysis, *J Power Sources* 257 (2014) 126–137, <https://doi.org/10.1016/j.jpowsour.2014.01.105>.
- [8] D. Werner, A. Loges, D.J. Becker, T. Wetzel, Thermal conductivity of li-ion batteries and their electrode configurations – a novel combination of modelling and experimental approach, *J Power Sources* 364 (2017) 72–83, <https://doi.org/10.1016/j.jpowsour.2017.07.105>.
- [9] R. Ponnappan, T.S. Ravigururajan, Contact thermal resistance of li-ion cell electrode stack, *J Power Sources* 129 (1) (2004) 7–13, <https://doi.org/10.1016/j.jpowsour.2003.11.006>.
- [10] V. Vishwakarma, C. Waghela, Z. Wei, R. Prasher, S.C. Nagpure, J. Li, F. Liu, C. Daniel, A. Jain, Heat transfer enhancement in a lithium-ion cell through improved material-level thermal transport, *J Power Sources* 300 (2015) 123–131, <https://doi.org/10.1016/j.jpowsour.2015.09.028>.
- [11] S.J. Bazinski, X. Wang, Experimental study on the influence of temperature and state-of-charge on the thermophysical properties of an lfp pouch cell, *J Power Sources* 293 (2015) 283–291, <https://doi.org/10.1016/j.jpowsour.2015.05.084>.
- [12] L. Sheng, L. Su, H. Zhang, Experimental determination on thermal parameters of prismatic lithium ion battery cells, *Int J Heat Mass Transf* 139 (2019) 231–239, <https://doi.org/10.1016/j.jheatmasstransfer.2019.04.143>.
- [13] A. Loges, S. Herberger, D. Werner, T. Wetzel, Thermal characterization of li-ion cell electrodes by photothermal deflection spectroscopy, *J Power Sources* 325 (2016) 104–115, <https://doi.org/10.1016/j.jpowsour.2016.05.082>.
- [14] T. Bandhauer, S. Garimella, T.F. Fuller, Electrochemical-thermal modeling to evaluate battery thermal management strategies: i. side cooling, *J Electrochem Soc* 162 (1) (2014) A125–A136, <https://doi.org/10.1149/2.0571501jes>.
- [15] W. Hesse, *Aluminium-Werkstoff-Datenblätter: Deutsch, 7th ed.*, Beuth Wissen, Beuth Verlag GmbH, Berlin, 2016.
- [16] M. Lewerenz, G. Fuchs, L. Becker, D.U. Sauer, Irreversible calendar aging and quantification of the reversible capacity loss caused by anode overhang, *Journal of Energy Storage* 18 (2018) 149–159, <https://doi.org/10.1016/j.est.2018.04.029>.
- [17] J.G. Hust, P.J. Giarratano, Thermal Conductivity and Electrical Resistivity Standard reference materials: austenitic stainless steel, SRM's 735 and 798, from 4 to 1200 (march 1975). NBS Special Publication 260-46 (1975).
- [18] *Springer handbook of materials measurement methods*, in: H. Czichos, T. Saito, L. Smith (Eds.), Springer handbooks, Springer, Berlin, 2006.
- [19] R.H. Bogaard, Thermal Conductivity of Selected Stainless Steels, in: T. Ashworth, D.R. Smith (Eds.), Thermal Conductivity 18, Springer Verlag, Boston, MA, 2012, pp. 175–185, https://doi.org/10.1007/978-1-4684-4916-7_20.
- [20] DIN EN 10088-3, Stainless steels - part 3: Technical delivery conditions for semi-finished products, bars, rods, wire, sections and bright products of corrosion resisting steels for general purposes; german version en 10088-3:2014, 2014, 10. 31030/2102108.
- [21] K.-Y. Oh, B.I. Epureanu, J.B. Siegel, A.G. Stefanopoulou, Phenomenological force and swelling models for rechargeable lithium-ion battery cells, *J Power Sources* 310 (2016) 118–129, <https://doi.org/10.1016/j.jpowsour.2016.01.103>.
- [22] A.L. Woodcraft, Predicting the thermal conductivity of aluminium alloys in the cryogenic to room temperature range, *Cryogenics* 45 (6) (2005) 421–431, <https://doi.org/10.1016/j.cryogenics.2005.02.003>.
- [23] C. Vargel, *Corrosion of Aluminium*, 1. ed., Elsevier, Amsterdam, 2004.
- [24] J. Sturm, A. Rheinfeld, I. Zilberman, F.B. Spingler, S. Kosch, F. Frie, A. Jossen, Modeling and simulation of inhomogeneities in a 18650 nickel-rich, silicon-graphite lithium-ion cell during fast charging, *J Power Sources* (412) (2019) 204–223.
- [25] F. Richter, S. Kjelstrup, P.J. Vie, O.S. Burheim, Thermal conductivity and internal temperature profiles of li-ion secondary batteries, *J Power Sources* 359 (2017) 592–600, <https://doi.org/10.1016/j.jpowsour.2017.05.045>.
- [26] R.W. Powell, C.Y. Ho, P.E. Liley, Thermal conductivity of selected materials, *NSRDS-NBS* 8 (1966).
- [27] O.S. Burheim, M.A. Onsrud, J.G. Pharoah, F. Vullum-Bruer, P.J.S. Vie, Thermal conductivity, heat sources and temperature profiles of li-ion batteries, *ECS Trans* 58 (48) (2014) 145–171, <https://doi.org/10.1149/05848.0145ecst>.
- [28] F. Richter, P.J. Vie, S. Kjelstrup, O.S. Burheim, Measurements of ageing and thermal conductivity in a secondary nmc-hard carbon li-ion battery and the impact on internal temperature profiles, *Electrochim. Acta* 250 (2017) 228–237, <https://doi.org/10.1016/j.electacta.2017.07.173>.
- [29] Y. Yang, X. Huang, Z. Cao, G. Chen, Thermally conductive separator with hierarchical nano/microstructures for improving thermal management of batteries, *Nano Energy* 22 (2016) 301–309, <https://doi.org/10.1016/j.nanoen.2016.01.026>.
- [30] D. Sauerteig, N. Hanselmann, A. Arzberger, H. Reinshagen, S. Ivanov, A. Bund, Electrochemical-mechanical coupled modeling and parameterization of swelling and ionic transport in lithium-ion batteries, *J Power Sources* 378 (2018) 235–247, <https://doi.org/10.1016/j.jpowsour.2017.12.044>.
- [31] D.R. Lide, *CRC Handbook of chemistry and physics, 2003–2004: A ready-reference book of chemical and physical data*, 84th ed., CRC Press, Boca Raton, Fla., 2003.
- [32] E.J. Cheng, K. Hong, N.J. Taylor, H. Choe, J. Wolfenstine, J. Sakamoto, Mechanical and physical properties of lini 0.33 mn 0.33 co 0.33 o 2 (nmc), *J Eur Ceram Soc* 37 (9) (2017) 3213–3217, <https://doi.org/10.1016/j.jeurceramsoc.2017.03.048>.
- [33] S. Yan, J. Deng, C. Bae, X. Xiao, Thermal expansion/shrinkage measurement of battery separators using a dynamic mechanical analyzer, *Polym Test* 71 (2018) 65–71, <https://doi.org/10.1016/j.polymertesting.2018.08.028>.
- [34] C. Bindra, V.A. Nalimova, D.E. Sklovsky, W.A. Kamitakahara, J.E. Fischer, Statics and dynamics of interlayer interactions in the dense high-pressure graphite compound lic2, *Physical Review B* 57 (9) (1998) 5182–5190, <https://doi.org/10.1103/PhysRevB.57.5182>.
- [35] A. Schilling, J. Schmitt, F. Dietrich, K. Dröder, Analyzing bending stresses on lithium-ion battery cathodes induced by the assembly process, *Energy Technology* 4 (12) (2016) 1502–1508, <https://doi.org/10.1002/ente.201600131>.
- [36] M. Bauer, M. Wachtler, H. Stöwe, J.V. Persson, M.A. Danzer, Understanding the dilation and dilation relaxation behavior of graphite-based lithium-ion cells, *J Power Sources* 317 (2016) 93–102, <https://doi.org/10.1016/j.jpowsour.2016.03.078>.
- [37] D. Sauerteig, S. Ivanov, H. Reinshagen, A. Bund, Reversible and irreversible dilation of lithium-ion battery electrodes investigated by in-situ dilatometry, *J Power Sources* 342 (2017) 939–946, <https://doi.org/10.1016/j.jpowsour.2016.12.121>.
- [38] Y. Ye, L.H. Saw, Y. Shi, K. Somasundaram, A.A. Tay, Effect of thermal contact resistances on fast charging of large format lithium ion batteries, *Electrochim. Acta* 134 (2014) 327–337, <https://doi.org/10.1016/j.electacta.2014.04.134>.
- [39] L. Chen, Z. Chen, S. Liu, B. Gao, J. Wang, Effects of particle size distribution on compacted density of lithium iron phosphate 18650 battery, *J Electrochem. Energy Convers. Storage* 15 (4) (2018) 2277, <https://doi.org/10.1115/1.4040825>.
- [40] P.-C. Tsai, B. Wen, M. Wolfman, M.-J. Choe, M.S. Pan, L. Su, K. Thornton, J. Cabana, Y.-M. Chiang, Single-particle measurements of electrochemical kinetics in NMC and NCA cathodes for li-ion batteries, *Energy Environ. Sci.* 11 (4) (2018) 860–871, <https://doi.org/10.1039/c8ee00001h>.
- [41] W.N. dos Santos, C.Y. Iguchi, R. Gregorio, Thermal properties of poly(vinylidene fluoride) in the temperature range from 25 to 210 °C, *Polym Test* 27 (2) (2008) 204–208, <https://doi.org/10.1016/j.polymertesting.2007.10.005>.
- [42] A. Gaitonde, A. Nimmagadda, A. Marconnet, Measurement of interfacial thermal conductance in lithium ion batteries, *J Power Sources* 343 (2017) 431–436, <https://doi.org/10.1016/j.jpowsour.2017.01.019>.
- [43] Compendium of Thermophysical Property Measurement Methods: vol. 2 Recommended Measurement Techniques and Practices, in: K.D. Maglič, A. Cezairliyan, V.E. Peletsky (Eds.), Springer US, Boston, MA and s.l., 1992, <https://doi.org/10.1007/978-1-4615-3286-6>.